

Quantitative Financial Risk Management: Assignment I, Market Risk

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by

Konstantinos Manos
(Student number: 2813642)

Konstantinos Flampouris
(Student number: 2759315)

Course Coordinator / Lecturer:
Dr. S.A. Borovkova

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Introduction

The purpose of this assignment is to construct a robust VaR system tailored to a diverse portfolio comprising both equity and fixed-income instruments. Our portfolio allocation strategy reflects a deliberate blend of assets, with 60% allocated to three major stock indices – the S&P 500, AEX, and DAX – and the remaining 40% invested in a bond linked to the Euribor 3-month rate. This allocation mix aims to strike a balance between potential returns from equity investments and the stability offered by fixed-income securities, while simultaneously diversifying risk exposures across different asset classes. Drawing upon data sourced from Yahoo Finance, we collected comprehensive information on the closing prices of the S&P 500, AEX, and DAX indices, as well as exchange rates from Euro to Dollar. It is noteworthy that while our portfolio is denominated in Euros, the S&P 500 index is priced in dollars, necessitating careful consideration of currency exchange fluctuations. Additionally, we obtained the Euribor 3-month rate from the Deutsche Bundesbank website. The dataset spans from April 10, 2012, to April 10, 2022, providing a robust historical foundation for our risk assessment endeavors.

$$R_{portfolio} = -[0.2LogR_{S\&P} + 0.2LogR_{DAX} + 0.2LogR_{AEX} + +0.4(0.0002 + 0.25\Delta R_{rates})]$$

Figure 1 depicts the daily log-returns of three stock market indices (S&P 500, AEX, DAX) denominated in Euros (EUR) alongside the daily EURIBOR rates changes over a 10-year period. The y-axis represents the daily return, and the x-axis represents time in years. The red line represents the S&P 500, the blue line represents the AEX, and the green line represents the DAX. All three indices exhibit volatility in daily returns, with fluctuations around zero.The purple line represents the EURIBOR (Euro Interbank Offered Rate). EURIBOR is a benchmark interest rate for short-term loans between Eurozone banks. Unlike the stock market indices, EURIBOR rates tend to be less volatile and may display a more persistent pattern over time, depending on monetary policy decisions.Table 1 shows for each day the daily log-return of the S&P 500, the AEX, the DAX, and the 3-month EURIBOR change rates. It also shows the daily return of the portfolio overall.

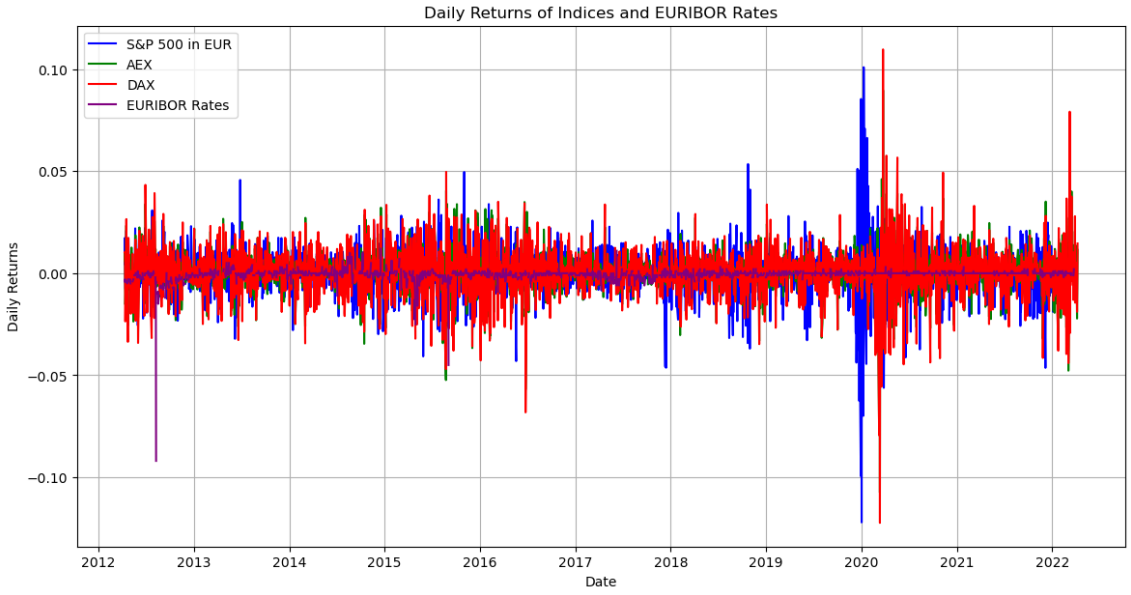


Figure 1: Daily returns

Table 1: Daily portfolio Returns

Date	S&P 500 in euro	DAX	AEX	EURIBOR rates	Portfolio Returns
2012-04-11	0.004551	0.010338	0.006080	-0.004	-0.003874
2012-04-12	0.017371	0.010264	0.010105	-0.003	-0.007328
2012-04-13	-0.007265	-0.023630	-0.015311	-0.004	0.009545
2012-04-16	-0.010552	0.006271	-0.000327	-0.004	0.001225

VaR and ES estimation with mulpitple methods

VaR is a statistical measure that quantifies the level of financial risk within a firm, portfolio, or position over a specific time frame. This metric is most commonly used by investment and commercial banks to determine the extent and occurrence probability of potential losses in their investment portfolios. ES, also known as Conditional Value at Risk (CVaR), provides a more comprehensive measure as it estimates the average loss in the worst $(100 - \alpha)\%$ of cases, where α is our confidence level. This metric accounts for the tail risk of the distribution, which is particularly important in financial contexts where the "black swan" events (rare but severe losses) can have substantial impacts.VaR and ES are pivotal in the construction of a robust risk management framework. They provide clear benchmarks for assessing the potential for extreme losses in our portfolio and are integral components for strategic planning, risk assessment, and regulatory compliance.

Variance-Covariance method based in normal distribution

In this section, we discuss the implementation and evaluation of the variance-covariance method using a normal distribution to estimate Value at Risk (VaR) and Expected Shortfall (ES) for the given portfolio.The variance-covariance method relies on the assumption that asset returns follow a multivariate normal distribution. Under this assumption, the VaR can be calculated using the formula:

$$\text{VaR} = \mu_p + z \times \sigma_p$$

Where:

- μ_p is the mean return of the portfolio.
- σ_p is the standard deviation of the portfolio return.
- z is the z-score corresponding to the desired confidence level.

Similarly, the ES can be calculated as:

$$\text{ES} = \mu_p + \sigma_p \times \phi(z) \div (1 - \alpha)$$

Where:

- α is the confidence level.
- $\phi(z)$ is the standard normal density function.

Figure 2 and Table 2 provide insightful representations of the risk metrics for our portfolio using the variance-covariance method based on a normal distribution. These estimates are crucial for understanding the potential losses that the portfolio could incur under normal market conditions.

Table 2: Examples of VaR and ES estimates in the sample

Date	VaR 97.5%	ES 97.5	VaR 99%	ES 99%
2012-04-11	0.00834	0.01151	0.01003	0.01302
2012-06-20	0.00748	0.00925	0.00890	0.01052
...	...	...	...	...
2020-12-29	0.00752	0.00748	0.00879	0.00862
2021-03-10	0.00747	0.00940	0.00891	0.01069

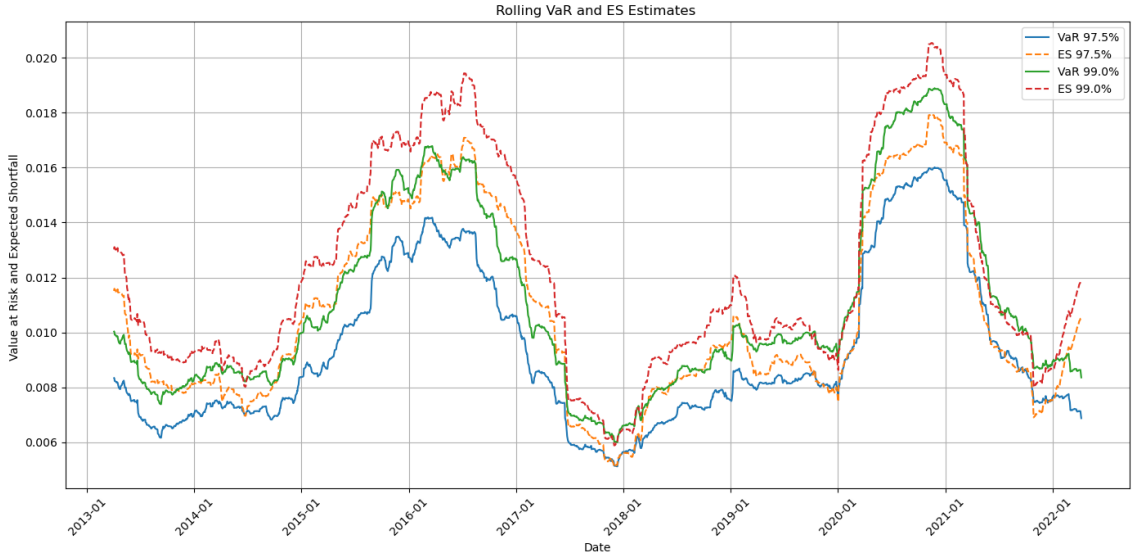


Figure 2: VaR and ES estimates with normal distribution

The VaR estimates presented in Figure 2 indicate the maximum expected loss at the 97.5% and 99% confidence levels over a rolling period from 2012 to 2022. For example, on April 11, 2012, the VaR at the 97.5% confidence level was 0.00834 (or 0.834%), which suggests that the portfolio is expected to lose no more than 0.834% of its value 97.5% of the time over the specified period. Such information is vital for risk management, as it helps to determine the necessary capital reserves to cover potential losses. In our data, ES captures more severe losses than VaR. For instance, still referring to April 11, 2012, the ES at a 97.5% confidence level was 0.01151 (or 1.151%), indicating that, on average, the worst 2.5% of cases could result in a loss exceeding 1.151% of the portfolio value.
Analyzing the time series data in Figure 2, we observe periods of elevated risk, notably around 2016 and 2020, where both VaR and ES metrics rise significantly. These spikes could correlate with market events such as economic downturns, elections, or other geopolitical events affecting market volatility.

Figure 3 below shows histograms and QQ plots for the daily returns of the SP 500, AEX, DAX, Euribor 3-month rate, and our daily portfolio return. As we can see in Figure 3, the histograms for the S&P 500, AEX, and DAX indices are bell-shaped, closely resembling the normal distribution, as evidenced by the overlay of the normal distribution curve. However, the tails, especially for S&P 500 and AEX, appear a bit fatter than those of a pure normal distribution, suggesting potential excess kurtosis (leptokurtosis), which indicates a higher probability of extreme returns than would be expected in a normal distribution. The histogram for the Euribor 3-month rate shows a distinct skew to the left, with a steep drop-off on the right side and a long left tail. This indicates a negative skew, meaning there are more extreme values on the negative side of the distribution. The distribution is clearly not symmetric, which is a departure from normality. In total, the histogram of the portfolio returns appears to have a bell shape, similar to the normal distribution, but with a slight rightward skew (positive skewness) suggesting that there are occasional larger-than-normal positive returns.

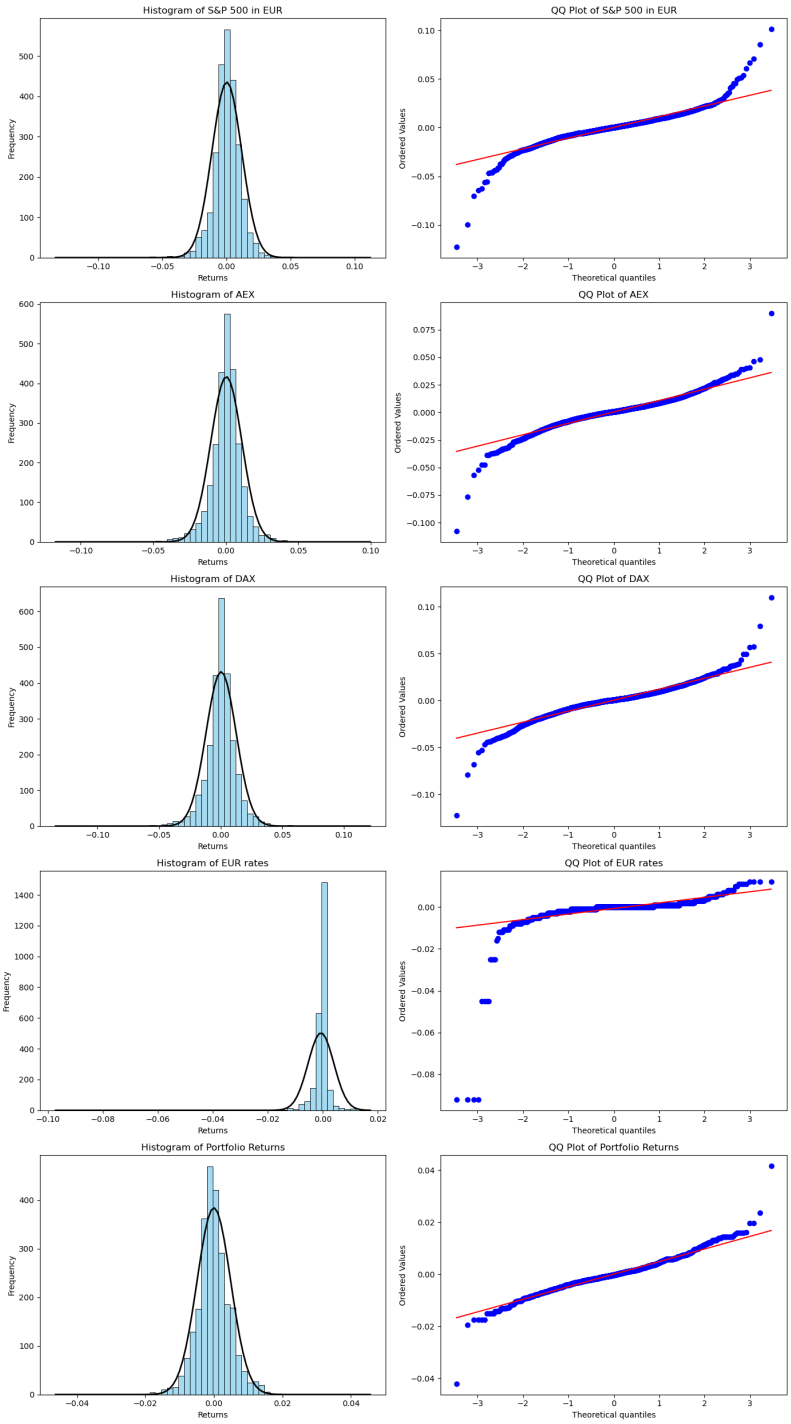


Figure 3: Histogram and QQ plot of each asset

In figure 3 we can see also QQ plots compare the quantiles of the sample data with the quantiles of a theoretical distribution (in this case, the normal distribution). The QQ plots for the S&P 500, AEX, and DAX show that the central parts of the distributions (between the quantiles of about -1 and 1) follow the line well, suggesting that the middle of the distribution is close to normal. However, the "S" shaped curve in the plots indicates heavier tails than the normal distribution, with more extreme values present in both the lower and upper tails. For the Euribor 3-month rate, the QQ plot, displays a pronounced deviation from the line in the lower tail, which again suggests the presence of negative skewness and potentially extreme negative returns that are not captured by a normal distribution. The QQ plot of the portfolio returns demonstrates a pattern similar to that of the individual indices, with a generally good fit along the line in the central quantiles but with deviations in the tails, especially in the upper tail, suggesting the presence of larger-than-expected positive returns.

The returns for the S&P 500, AEX, and DAX indices, as well as the aggregated portfolio, display characteristics of a normal distribution around the mean but have fatter tails than a normal distribution, which could be indicative of potential for extreme movements (either gains or losses). The Euribor 3-month rate shows significant non-normality, with a distribution skewed to the left, highlighting the potential for sudden and extreme decreases more than increases. While the middle range of returns may be well-modeled by normal distribution assumptions, the tails of the distributions for these financial instruments are not. This implies that risk models based solely on normal distributions may underestimate the probability and impact of extreme market movements.

Variance-Covariance method based in t-student distribution

The normal distribution has long been a staple in financial modeling due to its simplicity and mathematical tractability. However, empirical evidence suggests that financial asset returns often exhibit heavier tails than those assumed by the normal distribution. On the contrary The Student-t distribution offers a more flexible framework for capturing the characteristics of financial asset returns, particularly in its ability to accommodate heavier tails. By incorporating the Student-t distribution into the Variance-Covariance method, we aim to enhance the accuracy of risk estimation. Table 3 shows examples of our portfolio Value at Risk (VaR) and Expected Shortfall (ES) estimates for a t-student distribution. The table considers different degrees of freedom (Dof) and dates.

Table 3: Examples of VaR and ES estimates t-student distributrion

Date	VaR 97.5%	ES 97.5	VaR 99%	ES 99%
Degree of freedom (Dof)=3				
2013-06-10	0.009196509	0.017413939	0.012817563	0.024069666
2017-07-04	0.009062962	0.017125611	0.012615811	0.023655973
2022-02-24	0.007254348	0.014175704	0.010304281	0.019781672
Dof=4				
2013-06-10	0.009777663	0.015929458	0.012946467	0.020756125
2017-07-04	0.009633169	0.015669092	0.012742287	0.020404845
2022-02-24	0.007743841	0.012925359	0.010412853	0.016990752
Dof=5				
2013-06-10	0.00990659	0.014855552	0.012747781	0.018739738
2017-07-04	0.009759668	0.014615413	0.012547343	0.018426438
2022-02-24	0.007852433	0.012020832	0.010245504	0.015292395
Dof=6				
2013-06-10	0.009937671	0.014143412	0.012560835	0.017485684
2017-07-04	0.009790163	0.013916686	0.012363919	0.017196005
2022-02-24	0.007878612	0.011421013	0.010088044	0.014236133

Figure 4 portrays a complex set of rolling Value at Risk (VaR) and Expected Shortfall (ES) calculations for a ten-year period, leveraging a Student's t-distribution model with varying degrees of freedom (DoF) at 97.5% and 99% confidence levels. The application of the Student's t-distribution is particularly relevant in financial risk measurement due to its ability to capture heavier tails in return distributions than the normal distribution, often a characteristic of financial return data.

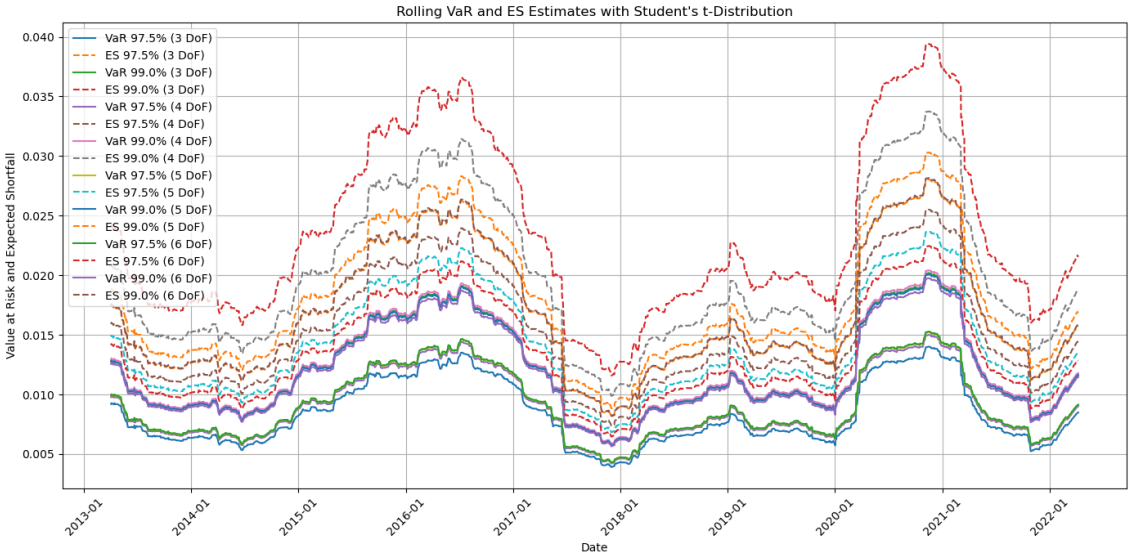


Figure 4: VaR and ES with Student's t-distribution

In Figure 4 we observe that, as the DoF decrease from 6 to 3, both VaR and ES estimates at corresponding confidence levels increase, reflecting a higher perceived risk. This is expected as lower DoF in a Student's t-distribution lead to heavier tails, indicating a greater likelihood of extreme losses. The lines representing 3 DoF are consistently at the top, demonstrating the highest risk estimates, whereas those for 6 DoF are at the bottom, showing the lowest risk estimates. This suggests that the return distributions are being recognized as having thicker tails with lower DoF. In addition, across all DoF and confidence levels, there are evident peaks and troughs corresponding to periods of high and low market volatility. Peaks, such as those around 2015–2016 and 2020, likely correspond to periods of financial stress or market downturns when asset volatility and the risk of loss were higher. The convergence of VaR and ES lines during these peaks suggests that as market conditions become more turbulent, the impact of changing the DoF becomes less distinct, possibly because actual returns distributions deviate significantly from the modeled t-distribution during such extreme conditions. Also, for both the 97.5% and 99% confidence levels, the ES is consistently higher than the VaR, which is in line with the definition of ES as a measure that accounts for the average loss in the worst cases beyond the VaR threshold. The gap between the VaR and ES lines also suggests the degree of loss severity beyond the VaR threshold. A wider gap indicates that when losses occur, they could be significantly larger than the VaR estimate.

Figure 4, underlines the importance of selecting appropriate DoF when using a Student's t-distribution to model financial returns. Risk managers must choose the DoF that best aligns with their risk assessment needs and the statistical properties of the assets in the portfolio. Given the variability in risk measures over time and across DoF, the choice of model parameters should not be static. Dynamic adjustments to DoF in response to changing market conditions might provide more accurate and responsive risk assessments. The observed peaks in VaR and ES estimates could serve as a signal for risk managers to review portfolio allocations and hedge positions, especially during times when the financial markets exhibit heightened volatility.

For the following, Figure 5 provides a collection of Quantile-Quantile (QQ) plots for the S&P 500, AEX, DAX indices, Euribor 3-month rates, and portfolio returns against a Student's t-distribution with varying degrees of freedom (DoF). QQ plots are a graphical tool to assess whether a set of observations matches a given distribution—in this case, the Student's t-distribution.

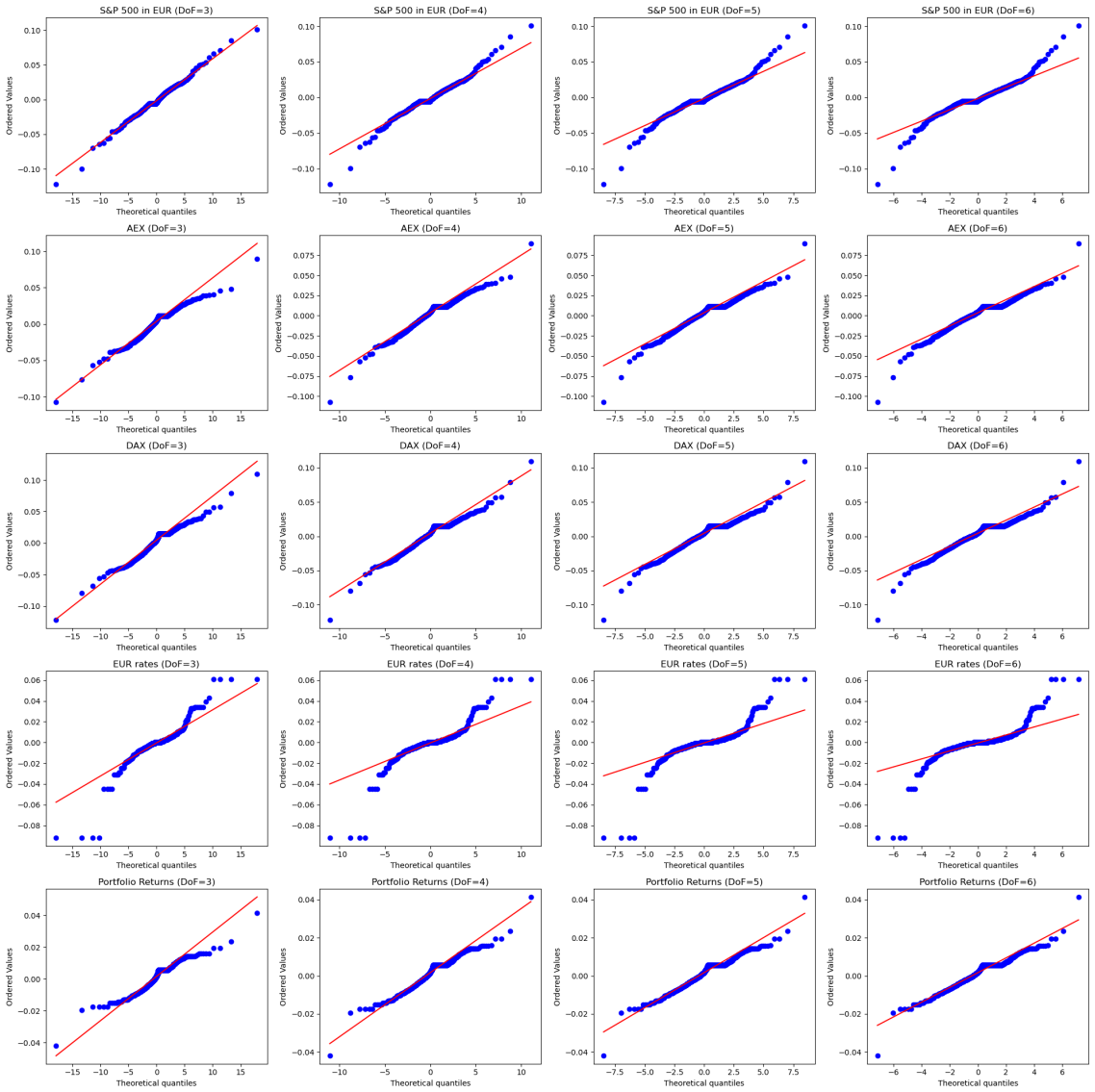


Figure 5: QQ plot of each asset for every degree of freedom

The S&P 500 QQ plots across all degrees of freedom indicate a reasonable match in the central quantiles. However, there is a noticeable deviation in the tails, especially the lower tail (left side of the plot), suggesting that extreme losses are more frequent than what the Student's t-distribution would predict. This tail behavior seems to reduce slightly with higher DoF, indicating that more DoF (which approaches a normal distribution) may offer a slightly better fit for the tails. Similar to the S&P 500, the AEX index QQ plots display good alignment in the middle of the distribution but deviations at both tails. As the degrees of freedom increase, the tails seem to adhere more closely to the line, suggesting that AEX returns have less extreme behavior than what is seen with lower degrees of freedom. The DAX index also shows deviations in the tails for lower DoF. However, the fit appears to improve as the DoF increase, especially for the right tail, which may imply that the DAX index has a less pronounced presence of extreme positive returns compared to the S&P 500 and AEX. In the other hand, the Euribor rates display a distinct pattern in their QQ plots. There is a consistent departure from the line in both tails across all DoF, indicating that extreme changes in Euribor rates are not well captured by the Student's t-distribution. This could reflect the nature of interest rate movements, which are often affected by policy decisions and macroeconomic factors, resulting in a return distribution that is not well modeled by simple parametric distributions. The portfolio returns QQ plots demonstrate a gradual improvement in fit with increasing DoF, particularly in the upper tail. This suggests that the portfolio, as an aggregate of various assets, may have a distribution that is closer to normal, but still benefits from the flexibility of a Student's t-distribution with more degrees of freedom to capture the tail risk.

Overall the plots indicate that while the Student's t-distribution with lower DoF provides a better fit for the center of the distribution, it often overestimates the thickness of the tails for the considered financial returns. As DoF increase, the fit for the tails generally improves, which might suggest that for these particular datasets, a Student's t-distribution with higher DoF, or potentially a different distribution altogether, might provide a better model for risk assessment purposes. For risk management, the choice of DoF when using the Student's t-distribution is non-trivial, as it has a significant impact on the assessment of tail risk. The QQ plots suggest that extreme market movements (represented by the tails) are not fully captured by lower DoF. Therefore, risk managers should consider using higher DoF or alternative models that better account for observed tail behavior to ensure robust risk assessment and management.

Historical simulation method

Historical simulation is a non-parametric approach to estimating Value at Risk (VaR) and Expected Shortfall (ES) for our portfolio. It relies on the assumption that the patterns of historical returns are indicative of future returns.

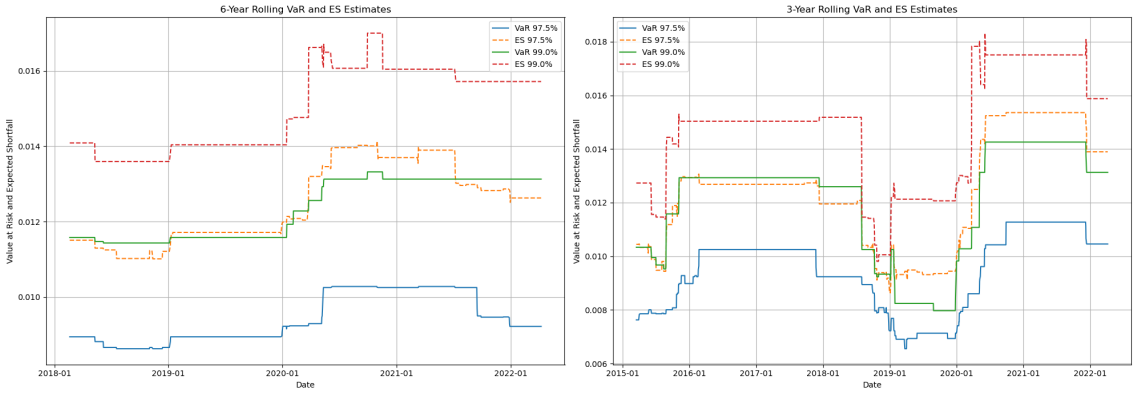


Figure 6: VaR and ES with Historical simulation method

Figure 6 presents two plots displaying the results of historical simulations used to calculate the Value at Risk (VaR) and Expected Shortfall (ES) for a portfolio using rolling windows of 3 years and 6 years. These simulations aim to estimate the risk of potential financial losses over specified periods, which is critical for effective risk management. The 6-year rolling estimates (left plot) show higher VaR and ES values compared to the 3-year estimates, which is indicative of capturing more extreme risk scenarios in the dataset. This is because the 6-year window includes a longer historical period, thus considering a wider range of market conditions, including periods of high volatility that might have occurred further in the past. The longer the historical window, the more likely it is to include outliers and extreme market events, resulting in higher risk estimates.In the other hand,the 3-year rolling estimates (right plot) are lower, suggesting that recent market conditions have been less volatile or that extreme losses were less pronounced or frequent during this shorter period. The 3-year window may respond more quickly to changes in market conditions, which can be beneficial for reflecting the current risk environment but may underestimate the risk if recent history has been unusually calm.

Table 4: Examples of VaR and ES estimates Historical simulation method

Date	VaR 97.5%	ES 97.5	VaR 99%	ES 99%
3 years				
2018-04-23	0.009236357	0.011952059	0.012594888	0.015182031
2022-03-04	0.010458718	0.013896049	0.013128569	0.015876168
6 years				
2018-04-23	0.008944682	0.011507486	0.0115803	0.014088123
2022-03-04	0.009220031	0.012627854	0.013128569	0.015717054

The stair-step pattern in both plots reflects the discrete adjustments in the risk levels as the rolling windows move forward in time. The changes in VaR and ES levels may correspond to the addition or dropping of historical data points that fall into or out of the rolling window. Sharp increases or drops in risk levels could indicate the entry or exit of periods of significant gains or losses from the rolling window. The information conveyed by the historical simulation approach is critical for risk managers as it illustrates the impact of considering different historical depths in risk assessment. While a longer look-back period provides a more comprehensive view of risk through different market cycles, it may also result in higher capital requirements to cover potential losses. Conversely, a shorter window may provide a more optimistic view of risk that aligns more closely with recent trends but could leave the portfolio underprepared for future market turmoil.

GARCH(1,1)

The constant conditional correlation (CCC) GARCH method is a widely used technique for estimating the Value at Risk (VaR) and Expected Shortfall (ES) of a portfolio. It combines aspects of GARCH models with a constant correlation matrix to capture the volatility and dependence between different assets in your portfolio. First we fit the GARCH(1,1) model:

$$\sigma_n^2 = \omega + \alpha u_{n-1}^2 + \beta \sigma_{n-1}^2$$

From our historical data we estimated the parameters of our model in Table 5

Table 5: Estimations of ω, α, β for GARCH(1,1)

	coefficients	std errors
ω	5.3535e-07	1.784e-13
α	0.2000	2.916e-08
β	0.7800	7.753e-03

Figure 7 illustrates the conditional volatility for our portfolio's returns, estimated using the Constant Conditional Correlation (CCC) GARCH(1,1) method. The x-axis represents time, likely ranging from 2012 to 2022. The y-axis represents the conditional volatility, which is a measure of the portfolio's riskiness at a given point in time. The conditional volatility fluctuates over time, indicating that the riskiness of the portfolio is not constant.There seems to be periods of higher volatility followed by periods of lower volatility.

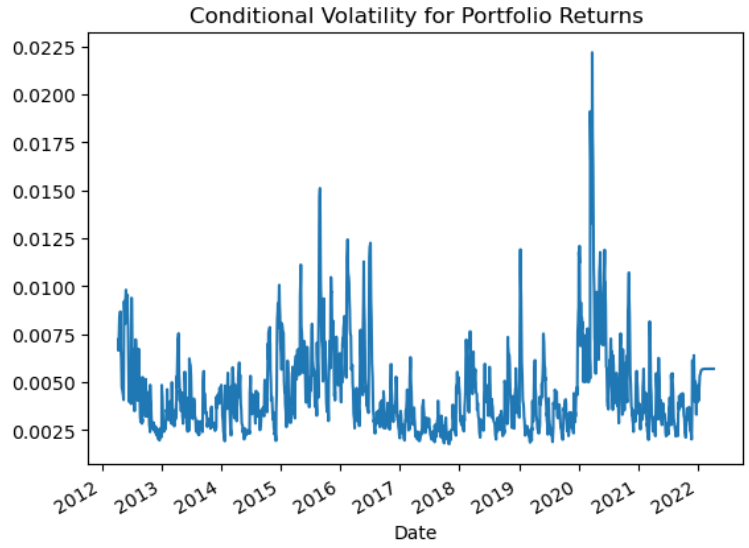


Figure 7: VaR and ES GARCH(1,1) method

We estimated Value at Risk (VaR) and Expected Shortfall (ES) using the Constant Conditional Correlation (CCC) GARCH(1,1) model. Table 6 presents the five first estimates on VaR and Expected Shortfall at two different confidence levels, 97.5% and 99%.

Table 6: Five first estimates of VaR and ES GARCH(1,1)

Date	VaR 97.5%	ES 97.5%	VaR 99%	ES 99%
2012-04-11	0.014120	0.016842	0.016760	0.019201
2012-04-12	0.013004	0.015511	0.015435	0.017683
2012-04-13	0.013237	0.015789	0.015711	0.018000
2012-04-16	0.014455	0.017242	0.017158	0.019657
2012-04-17	0.015339	0.018296	0.018206	0.020858

Figure 8 presents VaR and ES estimates at various confidence levels for individual stocks (SP 500, AEX, DAX), a Euribor 3-month bond, and our portfolio. For the three assets of the portfolio S&P 500, AEX, DAX the figure shows a significant increase in both VaR and ES estimates during certain periods, which may correspond to known financial crises or periods of market stress, such as the European debt crisis, the 2015–16 stock market selloff, or the 2020 coronavirus pandemic market crash. The spikes in VaR and ES are more pronounced at the 99% confidence level compared to the 97% level, indicating a larger estimation of risk as the confidence level increases. This is expected, as a higher confidence level corresponds to a more extreme percentile of the loss distribution. It is also observed that the ES estimates are consistently higher than the VaR for corresponding indices and confidence levels, which aligns with the nature of ES as a measure of the average loss in the worst-case scenarios beyond the VaR threshold. For Euribor rates the VaR and ES plots exhibit some of the highest spikes, particularly at the 99% confidence level. The magnitude of these spikes can be associated with the sensitivity of interest rates to major economic events and policy decisions by central banks. The extreme values in risk measures for Euribor rates, especially during the 2011–12 period, might reflect the market's reaction to the European Central Bank's monetary policy actions in response to the debt crisis. Overall, the portfolio returns plots show a risk profile that appears to be more contained compared to the individual indices, suggesting that the portfolio may be diversified across asset classes. However, it still experiences spikes in risk measures that correspond to the broader market events. The periods of heightened VaR and ES in the portfolio returns can be of particular interest to portfolio managers as they reflect times when a diversified portfolio might still face significant risks.

In conclusion, the VaR and ES time series captured in Figure 8 emphasize the dynamic nature of risk and the critical importance of maintaining vigilant risk management practices. These measures are essential for financial institutions in both setting aside appropriate capital for potential losses and for regulatory compliance. It also highlights that no market participant is immune to systemic events that cause market-wide stress, underscoring the need for continuous monitoring and updating of risk models to reflect the most current data.

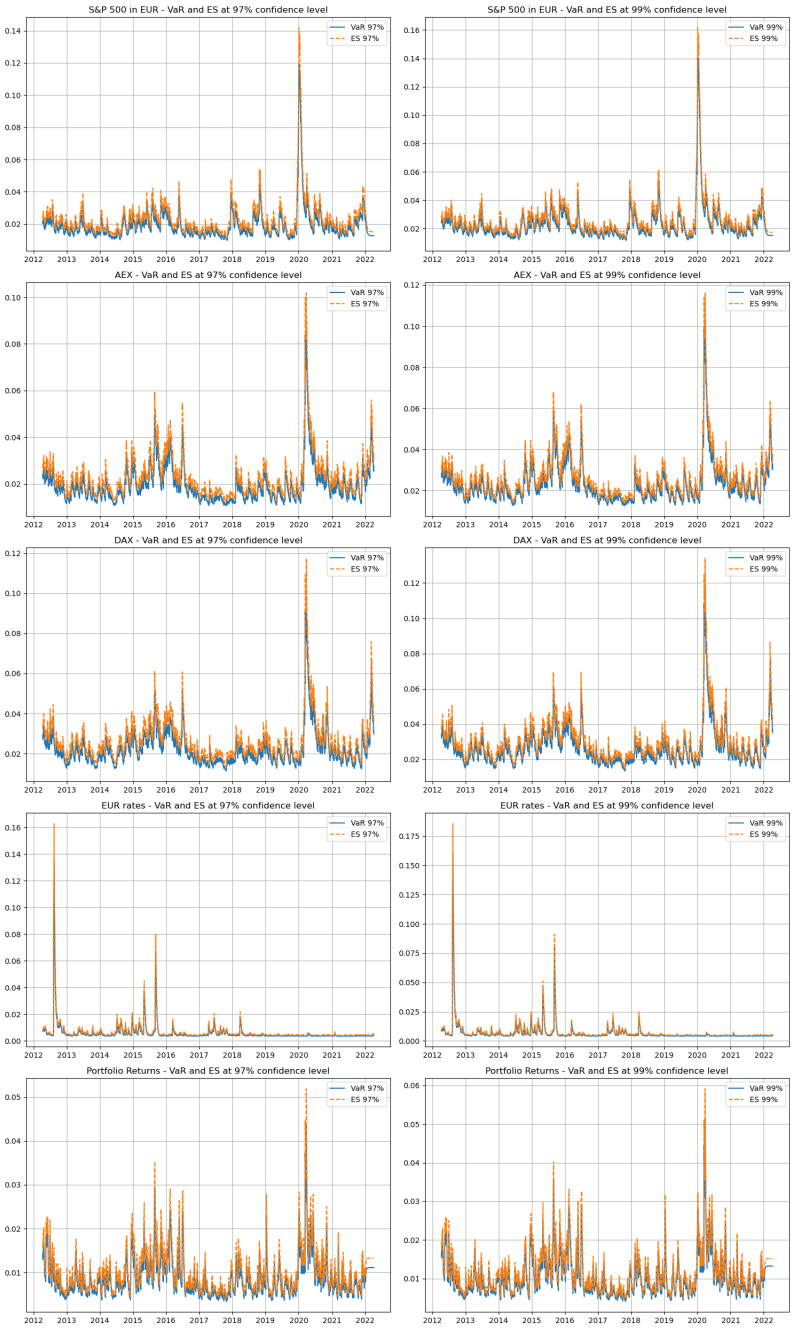


Figure 8: VaR and ES GARCH(1,1) method

Filtered Historical Simulation with EWMA

Filtered Historical Simulation (FHS) with EWMA involves applying the EWMA model to each risk factor in the portfolio. By calculating the volatility of each asset or index using EWMA, FHS incorporates these volatility estimates into the simulation process. This integration allows FHS to capture the dynamic nature of risk factors and adjust the simulated scenarios accordingly. In Table 7 we can see some examples, random dates, of VaR and ES estimated with Filtered Historical simulation method for confidence levels 97.5% and 99%.

Table 7: Examples VaR and ES, Filtered Historical simulation method

Date	VaR 97.5%	ES 97.5%	VaR 99%	ES 99&
2013-05-17	0.006622224	0.010581445	0.009964738	0.013400047
2015-07-08	0.008581186	0.01060861	0.011580356	0.012036302
2021-02-18	0.013128569	0.015933706	0.014258869	0.019817811
2022-03-31	0.006792612	0.008082823	0.008316627	0.010395134

Figure 9 showcases the Value at Risk (VaR) and Expected Shortfall (ES) for the S&P 500, AEX, DAX indices, Euribor rates, and portfolio returns, calculated using the filtered historical simulation approach with an Exponentially Weighted Moving Average (EWMA). This method adjusts for the most recent information to be more reflective of the current market conditions.

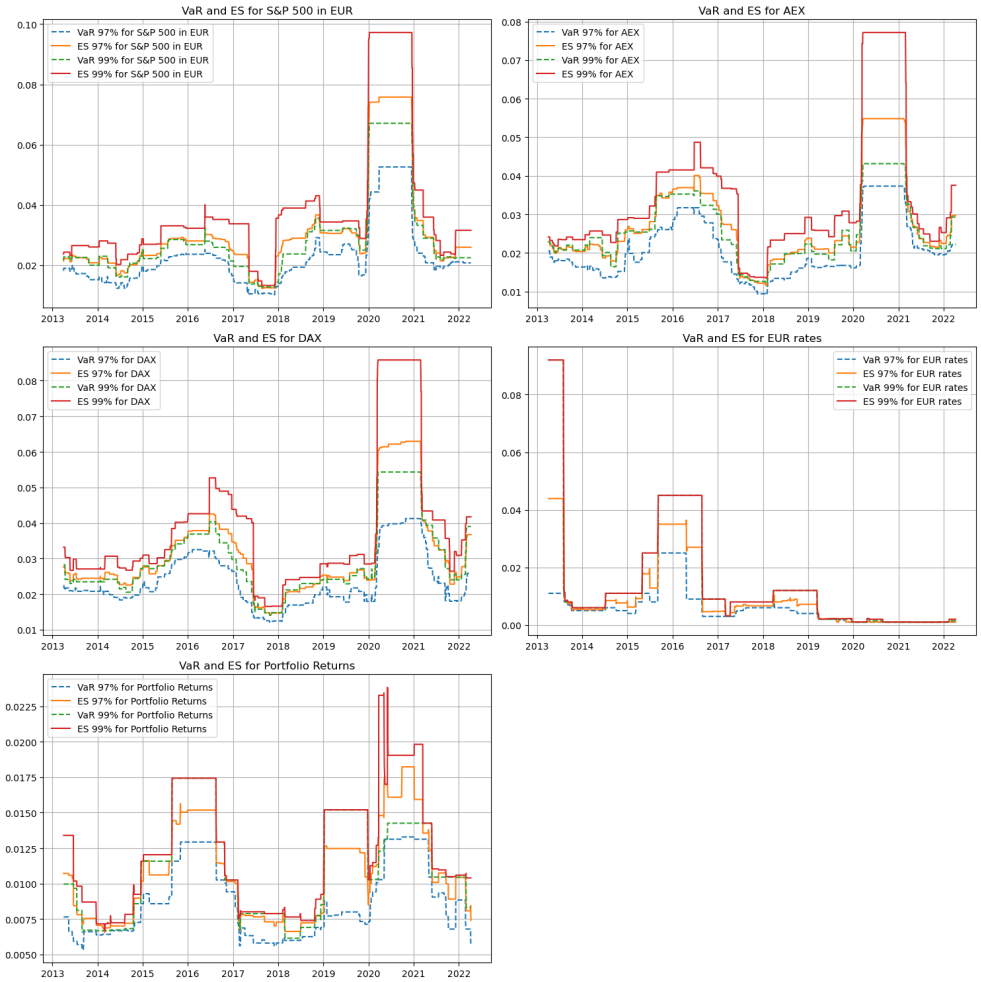


Figure 9: VaR and ES Filtered Historical Simulation method

The VaR and ES estimates for the S&P 500, AEX, and DAX show significant variability over the observed period, with notable peaks corresponding to periods of high market stress. The peaks are more pronounced in the ES estimates, particularly at the 99% confidence level, indicating a higher prediction of potential loss in adverse conditions. There are notable spikes in risk measures during periods such as the 2015–2016 market selloff and the 2020 pandemic-induced market crash, where extreme market volatility resulted in increased risk estimations. for Euribor rates are relatively stable compared to the equity indices, except for some isolated spikes. This could be reflective of infrequent but severe shifts in the interest rate environment, likely due to critical policy decisions or unexpected economic events.

Additionaly the portfolio returns demonstrate a risk profile with less extreme fluctuations than the individual equity indices. This might suggest that the portfolio benefits from diversification, although it is not completely insulated from systemic market shocks, as seen by the spikes in VaR and ES during market downturns. Similar to the individual assets, the ES measures for the portfolio are consistently higher than the VaR, indicating the portfolio's potential for experiencing greater losses in the worst-case scenarios compared to what the VaR predicts.

The application of EWMA likely provides more weight to recent observations, making the risk estimates more responsive to the latest market conditions. This can lead to more accurate risk estimations in fast-changing markets but may also result in more pronounced reactions to short-term volatility. The filtered historical simulation using EWMA takes into account the changing nature of volatility and could potentially offer a more precise estimation of VaR and ES compared to traditional historical simulation methods that treat all historical data equally. Figure 9 emphasizes the importance of incorporating recent market data into the risk measurement process and demonstrates that the filtered historical simulation method with EWMA is a valuable tool for capturing the dynamic nature of market risk. The elevated spikes in VaR and ES during periods of market turmoil underline the need for robust risk management systems that can adapt to rapid changes in the financial landscape.

Blacktest

Backtesting is a critical component in the evaluation and validation of Value at Risk (VaR) models, The primary objective of backtesting is to assess the accuracy and reliability of VaR estimates by comparing predicted outcomes against actual portfolio performance. Our analysis involves estimating VaR and Expected Shortfall (ES) using various methodologies and then subjecting these estimates to rigorous backtesting. This involves scrutinizing the number of VaR violations observed over a given period, comparing them with the expected number of violations, and assessing the average discrepancy between predicted and actual outcomes. Additionally, we examine the temporal distribution of VaR violations to discern whether they occur randomly or exhibit clustering, which could indicate underlying dependencies in the portfolio's risk structure.

We used Ad-hoc for statistical testing:
Let

$$I_t = 1_{\{L^{\Delta}_{t+1} > VaR^t_\alpha\}}$$

be the indicator of VaR violation on day $t+1$. Under the null hypothesis that VaR model is correct these indicators are Bernoulli random variables:

$$I_t = 1_{\{Z_{t+1} > q_\alpha(F_Z)\}} \sim Be(1-\alpha)$$

The total number of violations in the test set of size N has Binomial distribution under the null hypothesis:

$$\Sigma_{t=1}^{N} I_t \sim Bin(N, 1-\alpha)$$

To conclude whether the observed number of violations significantly differs from the expected number: N * (1-a), the binomial test statistic can be used:

$$T(N) = \frac{\Sigma_{t=1}^{N} I_t - N(1-\alpha)}{\sqrt{N\alpha(1-\alpha)}} \sim Normal(0,1)$$

and its p-value can be computed.

Table 8 presents the results of a backtest of Value at Risk (VaR) estimates for Variance-Covariance(Normal), GARCH(1,1) and Filtered Historical Simulation with EWMA over the 10 years-period.

Table 8: Blacktest of previous VaR systems

Year	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	All
Trading days	194	258	258	258	258	258	258	259	259	70	2330
Results for 97.5%											
Expected viol.	4.85	6.45	6.45	6.45	6.45	6.45	6.45	6.475	6.475	1.75	58.25
Var-Cov(Normal)	4 (0.7167)	10 (0.1156)	8 (0.3189)	0 (1.00)	7 (0.4664)	7 (0.4664)	8 (0.3189)	9 (0.2031)	2 (0.9891)	0 (1.00)	55
GARCH(1,1)	4 (0.7167)	4 (0.8877)	13 (0.0139)	3 (0.9571)	11 (0.0616)	7 (0.4664)	11 (0.0616)	9 (0.2031)	12 (0.0312)	0 (1.00)	74
EWMA	5 (0.5347)	20 (1e-5)	10 (0.1156)	0 (1.00)	6 (0.6263)	10 (0.1156)	9 (0.2001)	20 (1.1e-5)	2 (0.9891)	0 (1.00)	82
Results for 99%											
Expected viol.	1.95	2.58	2.58	2.58	2.58	2.58	2.58	2.59	2.59	0.7	23.3
Var-Cov(Normal)	0 (1.00)	6 (0.0467)	5 (0.1187)	0 (1.00)	1 (0.9252)	2 (0.7302)	8 (0.0048)	3 (0.4798)	1 (0.9259)	0 (1.00)	26
GARCH(1,1)	3 (0.3070)	2(0.7302)	5 (0.1187)	1 (0.9252)	4 (0.2591)	1 (0.9252)	5 (0.1187)	4 (0.2613)	9 (0.0013)	0 (1.00)	34
EWMA	1 (0.8576)	12 (1.5e-5)	5 (0.1187)	0 (1.00)	3 (0.4772)	6 (0.0467)	4 (0.2591)	9 (0.0013)	2 (0.7322)	0 (1.00)	42

Upon examination of the table, it is evident that the variance-covariance method consistently exhibits superior performance compared to alternative methodologies. Furthermore, the variance-covariance method demonstrates a closer alignment with the estimated number of violations when compared to the other methods.

For Varinace-Covariance with confidence level 97.5% in 2013, you observed 4 violations versus an expected 4.85. The p-value is 0.716768, which is not statistically significant, indicating that the model is performing as expected at this confidence level. For FHS wit confidence level97.5% in 2020, there were 20 violations when only about 6.475 were expected. The p-value is incredibly low, indicating that the model significantly underestimated risk and for GARCH with confidence level 99% in 2021, there were 9 violations with an expected number of 2.59, and the p-value is 0.001349. This is another case where the model significantly underestimated risk.

Figures 10 and 11 shows the Value at Risk (VaR) violations with confidence levels 97.5% and 99% respectively for three different models: Variance-Covariance, Filtered Historical Simulation with EWMA, and GARCH, plotted against time. There are three horizontal lines plotted across the entire graph, which represent the VaR thresholds for each model. There does seem to be some clustering of violations around 2020 and potentially in the 2015-2016 timeframe . While the violations are generally spread out, these periods show a higher concentration of points exceeding the VaR thresholds for some or all of the models.

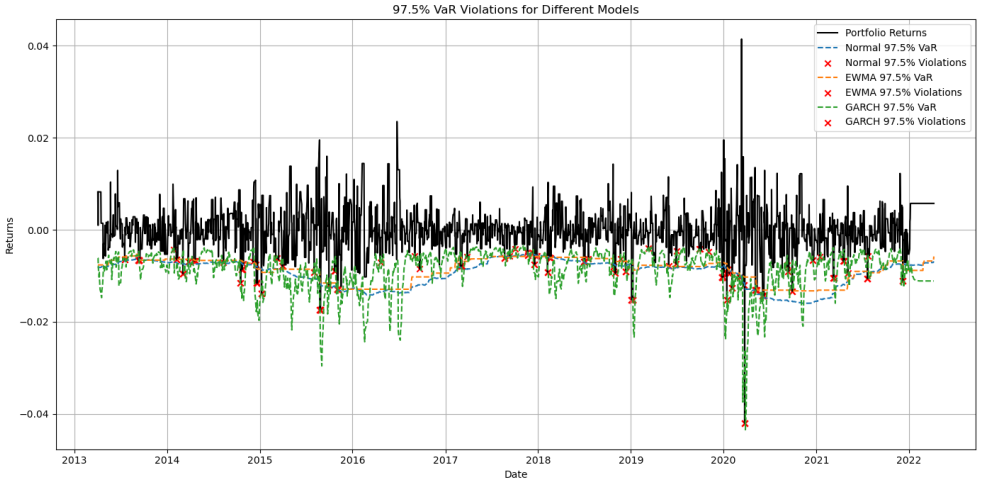


Figure 10: 97.5% VaR violations for different models

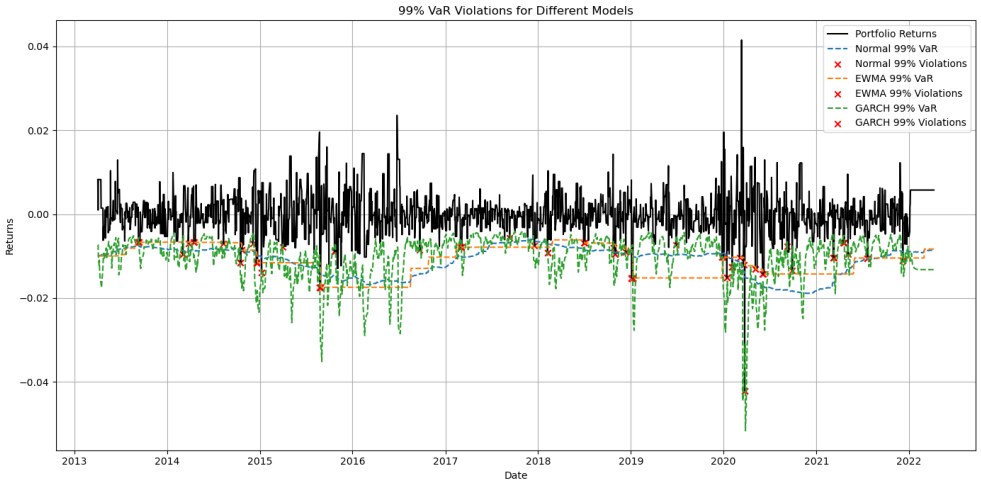


Figure 11: 99% VaR violations for different models

Comparison of VaRs using historical simulation

To compute the empirical 5-day and 10-day Value at Risk (VaR) for a financial portfolio, the historical simulation method is employed using non-overlapping returns data. This method involves aggregating portfolio returns into non-overlapping 5-day and 10-day periods, calculating the total return for each period. Specifically, for the 5-day VaR, 516 non-overlapping periods were used, and for the 10-day VaR, 258 non-overlapping periods were available, each calculated by summing returns from consecutive days without overlap between periods.The next step is to determine the VaR for each period at a specified confidence levels.

Simultaneously, the **Square Root of Time Rule** is applied to estimate VaR for these multi-day periods from the 1-day VaR. According to this rule, the n-day VaR can be approximated by multiplying the 1-day VaR by the square root of the number of days, $\sqrt{n}$ .This scaling approach assumes that the returns are normally distributed and exhibit a square root scaling in volatility. Hence, for a 5-day and a 10-day period, the 1-day VaR is scaled by $\sqrt{5}$ and $\sqrt{10}$, respectively.The procedure concludes by comparing the empirical VaRs, calculated directly from historical data, to those estimated using the Square Root of Time Rule. This comparison is crucial to assess the adequacy of the square root rule in capturing the risk profile of the portfolio over longer horizons, considering the actual return distribution and characteristics of the financial data.

Table 9: Comparisons of Value at Risk for different confidence levels

Method	Value at Risk
α=0.975	
1-Day VaR	0.009236
Empirical 5-Day VaR	0.032531
Square Root Time 5-Day VaR	0.020653
Empirical 10-Day VaR	0.049240
Square Root Time 10-Day VaR	0.029208
$\alpha = 0.99$	
1-Day VaR	0.012929
Empirical 5-Day VaR	0.036666
Square Root Time 5-Day VaR	0.028910
Empirical 10-Day VaR	0.061239
Square Root Time 10-Day VaR	0.040885

Table 9 above provides, detailed VaR estimates at two different confidence levels (α)—0.975 and 0.99—illustrating the expected maximum loss not to be exceeded with 97.5% and 99% certainty, respectively. At 97.5% confidence level for 1-Day VaR the baseline risk for a single day is 0.9236%. Empirical VaRs for 5 and 10 Days are considerably higher at 3.2531% and 4.9240%, respectively, suggesting a larger expected loss as the period extends.Square Root of Time Estimates yield lower estimates of 2.0653% for 5 days and 2.9208% for 10 days, indicating underestimation compared to empirical results. At 99% confidence levels 1-Day VaR increases to 1.2929%, reflecting higher risk perception. Empirical VaRs for 5 and 10 Days further elevated to 3.6666% and 6.1239%, underscoring more significant potential losses in higher confidence scenarios. Square Root of Time Estimates show 2.8910% for 5 days and 4.0885% for 10 days, again consistently lower than empirical calculations.

The comparison highlights that the Square Root of Time Rule generally provides conservative estimates relative to the empirical approach. This suggests that while the rule is useful for quick estimates, it might not fully capture the tail risk evident in empirical historical data, particularly under extreme market conditions. Consequently, relying solely on this rule could underestimate potential losses, particularly at higher confidence levels where understanding extreme outcomes is crucial. The empirical method, although more data-intensive, offers a nuanced view of risk reflective of actual market dynamics and historical data patterns.

Stress testing

In the realm of financial risk management, where uncertainty prevails as a constant, the ability to anticipate and mitigate extreme scenarios is paramount. While traditional risk measurement tools like Value at Risk (VaR) and Expected Shortfall (ES) offer valuable insights into potential losses under normal market conditions, they may not adequately capture the impact of severe, yet plausible, market shocks. In recognition of this limitation, stress testing emerges as a complementary tool to fortify risk assessment frameworks. By subjecting portfolios to extreme scenarios beyond the realm of routine fluctuations, stress testing provides a robust mechanism for evaluating resilience and identifying vulnerabilities in the face of adverse market conditions.

In the context of our portfolio, comprised of equity indices such as the SP 500, AEX, DAX, and a government bond with the Euribor 3-month rate, the implementation of stress testing takes on heightened significance. We adhere to established guidelines for stress testing, delineating specific scenarios tailored to the predominant risk factors within our portfolio. In accordance with these guidelines, we focus on two distinct scenarios: a 20% decline in equity index values or stock prices and a 2% downward shift in interest rates.

Utilizing the variance-covariance method, we have estimated the VaR under each of these stressed scenarios. This method, which relies on historical return data and covariance matrices, allows us to quantify the potential losses associated with extreme market movements. By applying this methodology to our portfolio, we aim to gain insights into the extent of potential downside risk and enhance our preparedness to navigate adverse market conditions.

Scenario 1: AEX ,DAX and SP 500 values decreasing by 20%

Figure 12 illustrates the Value at Risk (VaR) and Expected Shortfall (ES) estimates for a portfolio in normal market conditions, without the application of any stress scenario. (same with Figure 2, in section Variance-Covariance based in normal distribution). Figure 13 shows the VaR estimates following a severe market event—a 20drop in the equity indices. Notably, after the stress event is applied, the VaR estimates significantly increase, reflecting the heightened risk due to the assumed loss. We can see, that the increase in VaR indicates that the portfolio is more susceptible to larger financial losses in the event of a market downturn, emphasizing the importance of stress testing in risk management. Comparing the normal conditions to the stressed conditions, it is evident that the risk of loss becomes more pronounced when extreme market events are considered .

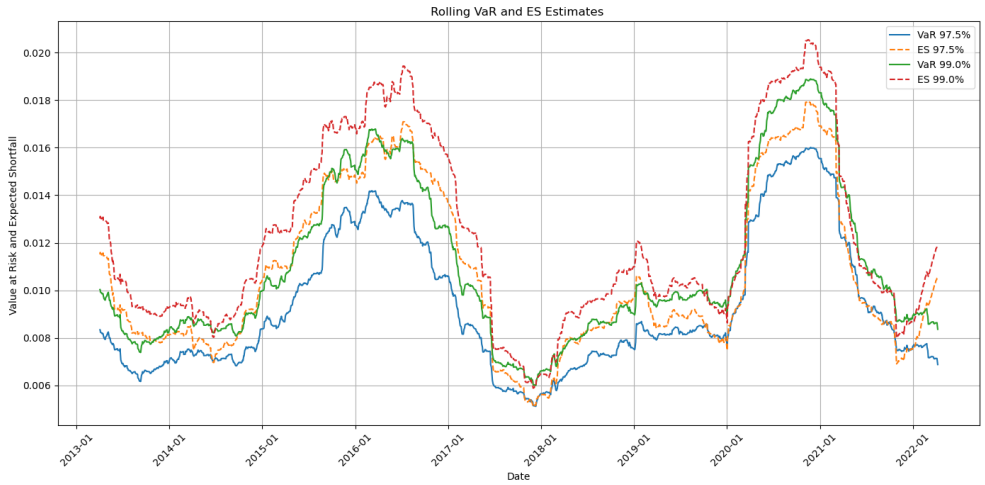


Figure 12: VaR and ES estimatations

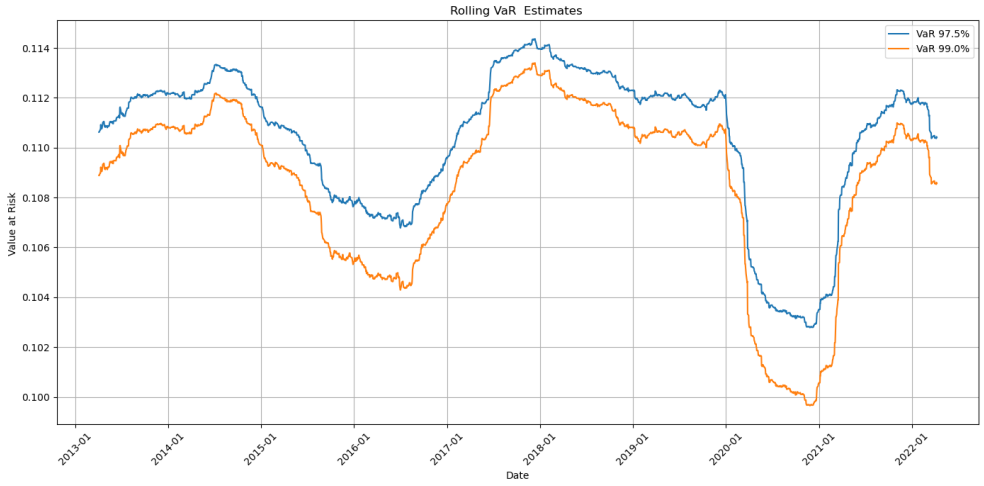


Figure 13: VaR estimations after the 20% decrease

The comparison between the two figures illustrates the importance of considering extreme but plausible market events in the risk management process.The stress test results highlight the need for effective risk mitigation strategies, such as diversification or hedging, to manage the potential for significant drawdowns in adverse market conditions. Observing the response of VaR under stress conditions also helps in understanding the non-linear behavior of risk in relation to market movements and can be vital for making informed decisions about risk appetite and asset allocation.

Scenario 2: Euribor 3-months rates increasing by 2%

Figure 12 shows the VaR and ES estimates for our portfolio without considering the 2% increase in the Euribor 3-month rate (see section Variance-Covariance based in Normal distribution). Figure 14 shows the VaR estimates for our portfolio after the 2% increase. Since the Euribor 3-month rate is a risk factor for our portfolio (increasing interest rates generally decrease bond prices), the VaR estimates in Figure 14 are higher than those in Figure 12. This indicates that a 2% increase in the Euribor 3-month rate would result in a greater potential loss for the portfolio.

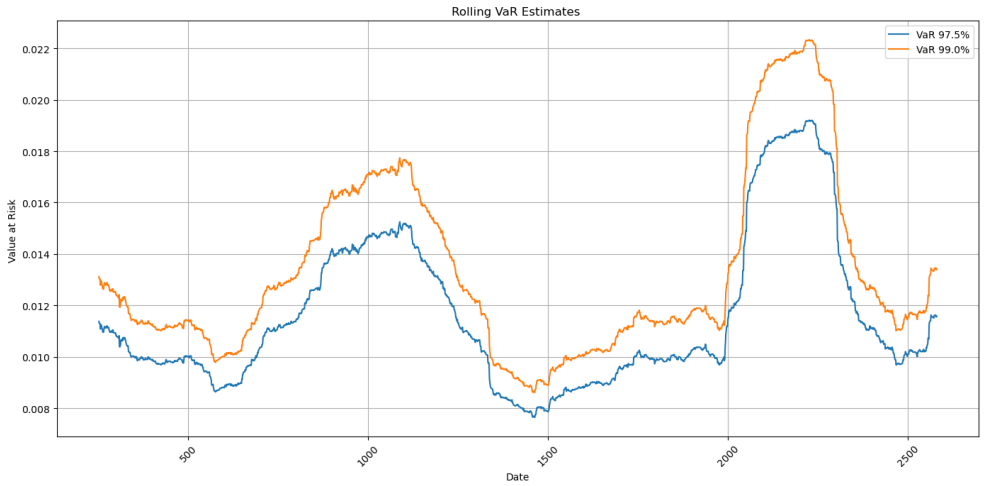


Figure 14: VaR estimations after the 2% increase

After the stress event, there is an evident increase in the VaR estimates, signifying that the portfolio is more at risk due to the increase in interest rates. Such a rise in Euribor rates can have a direct impact on the cost of borrowing and the value of interest-sensitive securities within the portfolio. The stress scenario results suggest the portfolio's vulnerability to interest rate risk and highlight the importance of managing interest rate exposure, especially in a rising rate environment.

The pre- and post-stress scenario VaR estimates emphasize the dynamic nature of financial risk and the value of considering potential adverse economic events in the risk management process. The marked increase in VaR following the hypothetical interest rate spike stresses the importance of preparing for interest rate fluctuations, and reinforces the need for continuous monitoring and adaptive risk management strategies to safeguard the portfolio's performance.

Conclusion

In conclusion, this comprehensive report delves into the construction of a robust Value at Risk (VaR) system tailored to a diverse portfolio comprising both equity and fixed-income instruments. Through meticulous data collection and analysis, various methodologies such as variance-covariance, filtered historical simulation with EWMA, and constant conditional correlation (CCC) GARCH were employed to estimate VaR and Expected Shortfall (ES) under different scenarios.

The report highlights the dynamic nature of financial risk, emphasizing the importance of adapting risk management frameworks to evolving market conditions. It underscores the limitations of simplistic models, such as the square root of time rule, in capturing the complexities of real-world risk dynamics, advocating for a nuanced approach that incorporates empirical data and stress testing.

Furthermore, the backtesting analysis provides valuable insights into the performance of different VaR models, aiding in the validation and refinement of risk estimation techniques, moreover statistical testing results showed us Variance- Covariance was the superior method for our portfolio . By subjecting the portfolio to stress scenarios, the report demonstrates the necessity of anticipating extreme market movements and fortifying risk management strategies accordingly.

Overall, the findings underscore the critical role of risk management in safeguarding portfolios against potential losses and navigating the uncertainties of financial markets. The report serves as a comprehensive guide for financial practitioners seeking to enhance their risk assessment capabilities and build resilience in the face of market volatility.